

Support Vector Machine Fundamentals

Part 1: Intuition, Hyperplane, Margin, and Hard-Margin SVM

CSE431: Machine Learning

Learning Objectives

By the end of this part, students will be able to:

- Explain the basic idea of Support Vector Machine.
- Understand why SVM is used for classification.
- Identify multiple possible decision boundaries.
- Explain margin and support vectors visually.
- Express SVM using hyperplane equations.
- Write the hard-margin SVM optimization problem.

What is Support Vector Machine?

Support Vector Machine (SVM) is a supervised Machine Learning algorithm.

It is mainly used for:

- Classification
- Regression

Core Idea

SVM finds the best decision boundary that separates classes with the maximum margin.

Why Do We Need SVM?

Suppose we have two classes:

Class A

and

Class B

Our goal is to separate them using a boundary.

But there may be many possible boundaries.

Important Question

Which boundary should we choose?

Classification Problem Example

Consider a simple student result prediction problem.

ID	Study Hours	Result
1	1	Fail
2	2	Fail
3	3	Fail
4	6	Pass
5	7	Pass
6	8	Pass

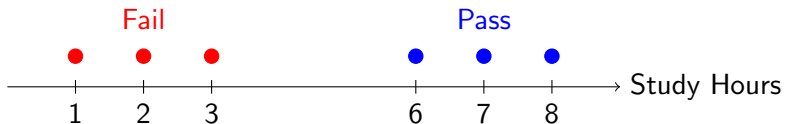
Goal:

Separate Fail and Pass students

Dataset Representation

Fail : 1, 2, 3

Pass : 6, 7, 8



Multiple Possible Decision Boundaries

Several boundaries can separate the two classes.

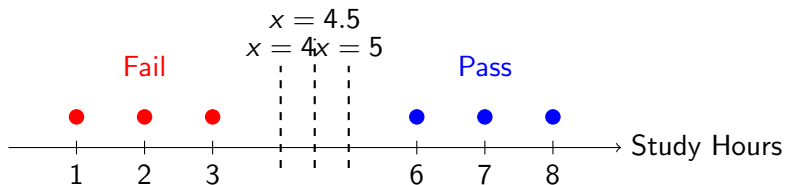
$$x = 4, \quad x = 4.5, \quad x = 5$$

All of them correctly separate Fail and Pass.

Question

If all boundaries classify correctly, which one is the best?

Visualizing Multiple Boundaries



All three boundaries separate the classes correctly.

Which Boundary Is Better?

Although all boundaries classify the training data correctly, they are not equally good.

A good boundary should:

- Separate the classes correctly.
- Stay far from the nearest data points.
- Generalize well to unseen data.

This distance from the boundary to the nearest points is called the **margin**.

Margin Concept

Margin is the distance between the decision boundary and the nearest data points.

Key Idea

A larger margin usually means better generalization.

SVM selects the boundary with the maximum margin.

Best Boundary = Maximum Margin Boundary

Boundary 1: $x = 4$

Nearest Fail point:

$$x = 3$$

Nearest Pass point:

$$x = 6$$

Distance from boundary:

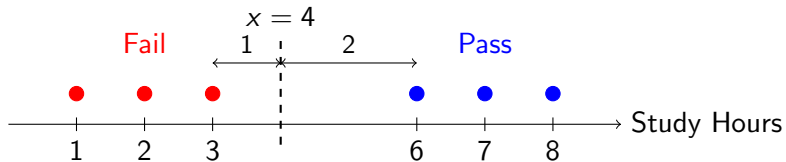
$$4 - 3 = 1$$

$$6 - 4 = 2$$

Margin:

$$\min(1, 2) = 1$$

Visualizing Boundary $x = 4$



$$\text{Margin} = \min(1, 2) = 1$$

Boundary 2: $x = 4.5$

Nearest Fail point:

$$x = 3$$

Nearest Pass point:

$$x = 6$$

Distance from boundary:

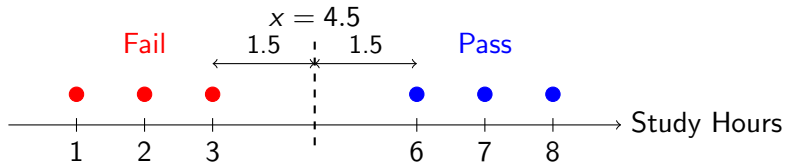
$$4.5 - 3 = 1.5$$

$$6 - 4.5 = 1.5$$

Margin:

$$\min(1.5, 1.5) = 1.5$$

Visualizing Boundary $x = 4.5$



$$\text{Margin} = 1.5$$

Boundary 3: $x = 5$

Nearest Fail point:

$$x = 3$$

Nearest Pass point:

$$x = 6$$

Distance from boundary:

$$5 - 3 = 2$$

$$6 - 5 = 1$$

Margin:

$$\min(2, 1) = 1$$

Comparing Margins

Boundary	Nearest Distances	Margin
$x = 4$	1, 2	1
$x = 4.5$	1.5, 1.5	1.5
$x = 5$	2, 1	1

Best boundary:

$$x = 4.5$$

because it has the largest margin.

Maximum Margin Classifier

SVM is also called a **Maximum Margin Classifier**.

It chooses the decision boundary that:

- Correctly separates the classes.
- Maximizes the distance from the nearest data points.
- Reduces the risk of poor generalization.

SVM chooses the boundary with maximum margin.

What Are Support Vectors?

Support Vectors are the closest data points to the decision boundary.

In our example:

$$\textit{Fail} : x = 3$$

$$\textit{Pass} : x = 6$$

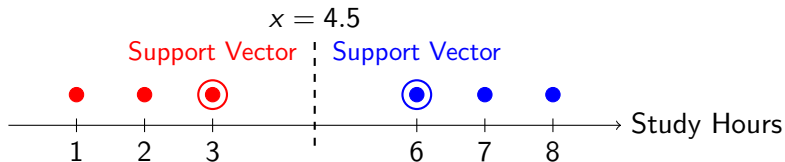
These two points determine the optimal boundary:

$$x = 4.5$$

Important

Support vectors are the most important points in SVM.

Support Vectors Visualization



Support vectors are the nearest points to the boundary.

Why Support Vectors Matter

Support vectors control the position of the decision boundary.

If we remove a non-support vector:

$$x = 1, \quad x = 2, \quad x = 7, \quad x = 8$$

the boundary usually does not change.

But if we remove a support vector:

$$x = 3$$

or

$$x = 6$$

the boundary may change.

From Visual Idea to Mathematics

So far, we learned visually:

- SVM finds the best boundary.
- The best boundary has maximum margin.
- Support vectors are closest to the boundary.

Now we express these ideas mathematically.

Next Step

Decision boundary becomes a hyperplane equation.

What is a Hyperplane?

In SVM, the decision boundary is called a **hyperplane**.

- In 1D: decision boundary is a point.
- In 2D: decision boundary is a line.
- In 3D: decision boundary is a plane.
- In higher dimensions: decision boundary is a hyperplane.

$$w^T x + b = 0$$

Hyperplane Equation

The hyperplane equation is:

$$w^T x + b = 0$$

where:

- x = input feature vector
- w = weight vector
- b = bias

For two features:

$$w_1 x_1 + w_2 x_2 + b = 0$$

Simple 1D Hyperplane Example

From our previous example:

Fail : 1, 2, 3

Pass : 6, 7, 8

Best boundary:

$$x = 4.5$$

This can be written as:

$$x - 4.5 = 0$$

So:

$$w = 1, \quad b = -4.5$$

Decision Function

SVM uses the decision function:

$$f(x) = w^T x + b$$

For the boundary:

$$x - 4.5 = 0$$

Decision function:

$$f(x) = x - 4.5$$

Classification is based on the sign of $f(x)$.

Classification Rule

$$f(x) > 0 \Rightarrow \text{Positive Class}$$

$$f(x) < 0 \Rightarrow \text{Negative Class}$$

$$f(x) = 0 \Rightarrow \text{On the Boundary}$$

For our example:

$$f(x) = x - 4.5$$

Prediction Examples

For $x = 7$:

$$f(7) = 7 - 4.5 = 2.5$$

$$f(7) > 0 \Rightarrow \textit{Pass}$$

For $x = 2$:

$$f(2) = 2 - 4.5 = -2.5$$

$$f(2) < 0 \Rightarrow \textit{Fail}$$

2D Hyperplane Example

Suppose we have two features:

$$x_1 = \text{Study Hours}$$

$$x_2 = \text{Attendance}$$

Hyperplane:

$$2x_1 + x_2 - 10 = 0$$

Here:

$$w = (2, 1), \quad b = -10$$

Margin Boundaries

For SVM, support vectors lie on the margin boundaries:

$$w^T x + b = 1$$

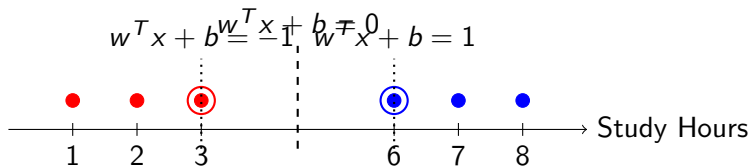
and

$$w^T x + b = -1$$

The decision boundary lies between them:

$$w^T x + b = 0$$

Visual Summary: Boundary and Margins



Correct Classification Condition

For a positive sample:

$$y_i = +1$$

we want:

$$w^T x_i + b \geq 1$$

For a negative sample:

$$y_i = -1$$

we want:

$$w^T x_i + b \leq -1$$

Unified SVM Constraint

Both conditions can be written together as:

$$y_i(w^T x_i + b) \geq 1$$

Class	Required Condition	Unified Form
$y_i = +1$	$w^T x_i + b \geq 1$	$y_i(w^T x_i + b) \geq 1$
$y_i = -1$	$w^T x_i + b \leq -1$	$y_i(w^T x_i + b) \geq 1$

Distance from Hyperplane

Distance of point x_i from the hyperplane

$$w^T x + b = 0$$

is:

$$\frac{|w^T x_i + b|}{\|w\|}$$

For support vectors:

$$|w^T x_i + b| = 1$$

Therefore:

$$\text{Distance from boundary to margin} = \frac{1}{\|w\|}$$

Total Margin Width

There are two margin boundaries:

$$w^T x + b = 1$$

and

$$w^T x + b = -1$$

Distance from decision boundary to each margin boundary:

$$\frac{1}{\|w\|}$$

So total margin width is:

$$\boxed{\frac{2}{\|w\|}}$$

Optimization Goal

SVM wants to maximize the margin:

$$\max \frac{2}{\|w\|}$$

This is equivalent to minimizing:

$$\|w\|$$

For mathematical convenience, we minimize:

$$\frac{1}{2} \|w\|^2$$

Reason

The square removes the square root and makes differentiation easier.

Hard-Margin SVM Optimization Problem

The hard-margin SVM problem is:

$$\min_{w,b} \frac{1}{2} \|w\|^2$$

subject to:

$$y_i(w^T x_i + b) \geq 1$$

for all training samples.

This is a constrained optimization problem.

Hard-Margin Assumption

Hard-margin SVM assumes the data is perfectly linearly separable.
That means there exists a hyperplane such that:

$$y_i(w^T x_i + b) \geq 1$$

for every training point.

Limitation

In real-world datasets, perfect separation is often not possible.

Hard Margin vs Soft Margin

Topic	Hard Margin SVM	Soft Margin SVM
Data assumption	Perfectly separable	May overlap
Violations	Not allowed	Allowed with penalty
Outlier handling	Sensitive	More flexible
Main use	Ideal data	Real-world data

Key Equations So Far

Hyperplane:

$$w^T x + b = 0$$

Prediction rule:

$$\hat{y} = \text{sign}(w^T x + b)$$

SVM constraint:

$$y_i(w^T x_i + b) \geq 1$$

Margin width:

$$\frac{2}{\|w\|}$$

Optimization problem:

Why We Need Dual Form Next

The hard-margin SVM problem is:

$$\min_{w,b} \frac{1}{2} \|w\|^2$$

subject to:

$$y_i(w^T x_i + b) \geq 1$$

This is a constrained optimization problem.

To solve it systematically, we introduce:

$$\alpha_i$$

called **Lagrange multipliers**.

Part 1 Summary

In this part, we learned:

- SVM finds a maximum-margin separating boundary.
- The closest points are called support vectors.
- The decision boundary is $w^T x + b = 0$.
- Margin boundaries are $w^T x + b = 1$ and $w^T x + b = -1$.
- Margin width is $\frac{2}{\|w\|}$.
- Hard-margin SVM solves a constrained optimization problem.

Next: **Dual Form Derivation**

Part 2

Dual Form Derivation of SVM

Review: Hard-Margin SVM

Primal optimization problem:

$$\min_{w,b} \frac{1}{2} \|w\|^2$$

subject to

$$y_i(w^T x_i + b) \geq 1$$

for all training samples.

Goal:

- Maximize margin
- Correctly classify all samples
- Find optimal w and b

Why Do We Need Dual Form?

The primal problem contains constraints:

$$y_i(w^T x_i + b) \geq 1$$

Directly solving constrained optimization problems can be difficult.

To simplify the problem, we use:

Lagrange Multipliers

A mathematical technique that converts a constrained optimization problem into an unconstrained one.

Lagrange Multipliers

For every constraint,

$$y_i(w^T x_i + b) - 1 \geq 0$$

introduce a multiplier

$$\alpha_i \geq 0$$

Each training sample gets its own multiplier.

$$\alpha_1, \alpha_2, \dots, \alpha_n$$

Constructing the Lagrangian

The primal objective:

$$\frac{1}{2} \|w\|^2$$

Constraint:

$$y_i(w^T x_i + b) - 1 \geq 0$$

Lagrangian:

$$L(w, b, \alpha) = \frac{1}{2} \|w\|^2 - \sum_{i=1}^n \alpha_i [y_i(w^T x_i + b) - 1]$$

Derivative with Respect to w

Take derivative:

$$\frac{\partial L}{\partial w} = w - \sum_{i=1}^n \alpha_i y_i x_i$$

Set equal to zero:

$$\frac{\partial L}{\partial w} = 0$$

Therefore,

$$w = \sum_{i=1}^n \alpha_i y_i x_i$$

Important Result

The weight vector is a linear combination of training samples.

Derivative with Respect to b

Take derivative:

$$\frac{\partial L}{\partial b} = - \sum_{i=1}^n \alpha_i y_i$$

Set equal to zero:

$$\frac{\partial L}{\partial b} = 0$$

Hence

$$\sum_{i=1}^n \alpha_i y_i = 0$$

Karush-Kuhn-Tucker (KKT)

For SVM:

$$\alpha_i \geq 0$$

$$y_i(w^T x_i + b) - 1 \geq 0$$

$$\alpha_i (y_i(w^T x_i + b) - 1) = 0$$

This means:

- Either $\alpha_i = 0$
- Or the point lies exactly on the margin

Key Takeaway

Non-zero Alpha $\implies$ Support Vector

Substituting Back

From earlier:

$$w = \sum_i \alpha_i y_i x_i$$

Substitute this into the Lagrangian.

After simplification:

$$W(\alpha) = \sum_i \alpha_i - \frac{1}{2} \sum_i \sum_j \alpha_i \alpha_j y_i y_j (x_i^T x_j)$$

Dual Optimization Problem

Maximize:

$$W(\alpha) = \sum_i \alpha_i - \frac{1}{2} \sum_i \sum_j \alpha_i \alpha_j y_i y_j (x_i^T x_j)$$

Subject to:

$$\alpha_i \geq 0$$

$$\sum_i \alpha_i y_i = 0$$

Interpretation of Alpha Values

After optimization:

- Most $\alpha_i = 0$
- Only a few $\alpha_i > 0$

Points with

$$\alpha_i > 0$$

become support vectors.

These points determine the final decision boundary.

Why Dual Form is Important?

Advantages:

- Automatically identifies support vectors
- Often easier to solve
- Enables Kernel Methods

Notice:

$$x_i^T x_j$$

appears everywhere.

Later we replace it with:

$$K(x_i, x_j)$$

which leads to Kernel SVM.

Part 2 Summary

- Introduced Lagrange multipliers
- Constructed the SVM Lagrangian
- Derived expressions for w and b
- Obtained the dual optimization problem
- Learned that non-zero alphas identify support vectors
- Prepared for solving a complete numerical example

Next: Full Step-by-Step Dual Form Numerical Example

Part 3

Complete Dual Form Numerical Example

Training Dataset

Point	Coordinates	Class
A	(2,2)	+1
B	(4,4)	+1
C	(2,0)	-1

Labels:

$$y_1 = +1, \quad y_2 = +1, \quad y_3 = -1$$

Dual Optimization Problem

We need to maximize:

$$W(\alpha) = \sum_i \alpha_i - \frac{1}{2} \sum_i \sum_j \alpha_i \alpha_j y_i y_j (x_i^T x_j)$$

Subject to:

$$\alpha_i \geq 0$$

$$\sum_i \alpha_i y_i = 0$$

What is a Dot Product?

In the dual form of SVM, we frequently compute:

$$x_i^T x_j$$

This is called the **dot product**.

For two vectors:

$$(a, b) \text{ and } (c, d)$$

the dot product is:

$$(a, b)^T (c, d) = ac + bd$$

Example:

$$(2, 3)^T (4, 5) = 2 \times 4 + 3 \times 5 = 8 + 15 = 23$$

Step 1: Computing the Dot Products

Training points:

$$x_1 = (2, 2)$$

$$x_2 = (4, 4)$$

$$x_3 = (2, 0)$$

Example 1:

$$x_1^T x_1 = (2, 2)^T (2, 2)$$

$$= 2 \times 2 + 2 \times 2$$

$$= 8$$

Example 2:

$$x_1^T x_2 = (2, 2)^T (4, 4)$$

$$= 2 \times 4 + 2 \times 4 = 16$$

Similarly,

$$x_1^T x_3 = (2, 2)^T (2, 0) = 4$$

$$x_2^T x_2 = (4, 4)^T (4, 4) = 32$$

$$x_2^T x_3 = (4, 4)^T (2, 0) = 8$$

$$x_3^T x_3 = (2, 0)^T (2, 0) = 4$$

Building the Gram Matrix

Putting all dot products together:

$$K = \begin{bmatrix} 8 & 16 & 4 \\ 16 & 32 & 8 \\ 4 & 8 & 4 \end{bmatrix}$$

where

$$K_{ij} = x_i^T x_j$$

Constraint Equation

Using

$$\sum_i \alpha_i y_i = 0$$

we get

$$\alpha_1 + \alpha_2 - \alpha_3 = 0$$

Thus

$$\alpha_3 = \alpha_1 + \alpha_2$$

Identifying Support Vectors

From the geometry of the dataset:

- Point A is closest to the opposite class.
- Point C is closest to the opposite class.
- Point B is much farther from the boundary.

According to the KKT conditions:

$$\alpha_i > 0$$

only for support vectors.

Therefore:

$$\alpha_2 = 0$$

and A and C become support vectors.

Building the Dual Objective

From the constraint:

$$\alpha_1 = \alpha_3 = \alpha$$

and

$$\alpha_2 = 0$$

The dual objective is:

$$W(\alpha) = \sum_i \alpha_i - \frac{1}{2} \sum_i \sum_j \alpha_i \alpha_j y_i y_j K_{ij}$$

Linear term:

$$\alpha_1 + \alpha_3 = \alpha + \alpha = 2\alpha$$

Computing the Quadratic Term

Recall:

$$\alpha_1 = \alpha_3 = \alpha$$

$$\alpha_2 = 0$$

Therefore, only points 1 and 3 contribute to the quadratic term.

The quadratic part is:

$$\sum_i \sum_j \alpha_i \alpha_j y_i y_j K_{ij}$$

Expanding:

$$= \alpha_1^2 y_1^2 K_{11} + \alpha_1 \alpha_3 y_1 y_3 K_{13} + \alpha_3 \alpha_1 y_3 y_1 K_{31} + \alpha_3^2 y_3^2 K_{33}$$

Expanding the Quadratic Term

Only support vectors A and C remain:

$$\alpha_1 = \alpha_3 = \alpha$$

From the Gram Matrix:

$$K_{11} = 8$$

$$K_{33} = 4$$

$$K_{13} = K_{31} = 4$$

Quadratic term:

$$= \underbrace{\alpha^2(8)}_{K_{11}} + \underbrace{\alpha^2(4)}_{K_{33}} - \underbrace{\alpha^2(4)}_{K_{13}} - \underbrace{\alpha^2(4)}_{K_{31}}$$

Since the two cross terms are identical:

$$= 8\alpha^2 + 4\alpha^2 - 2(4)\alpha^2$$

$$= 8\alpha^2 + 4\alpha^2 - 8\alpha^2$$

$$= 4\alpha^2$$

Final Dual Objective

Linear term:

$$2\alpha$$

Quadratic term:

$$4\alpha^2$$

Substitute into the dual objective:

$$\begin{aligned} W(\alpha) &= 2\alpha - \frac{1}{2}(4\alpha^2) \\ &= 2\alpha - 2\alpha^2 \end{aligned}$$

Now we maximize this function.

Maximizing the Objective Function

Dual objective:

$$W(\alpha) = 2\alpha - 2\alpha^2$$

Differentiate:

$$\frac{dW}{d\alpha} = 2 - 4\alpha$$

Set derivative equal to zero:

$$2 - 4\alpha = 0$$

$$\alpha = \frac{1}{2}$$

Therefore,

$\alpha = 0.5$

Optimal Alpha Values

$$\alpha_1 = 0.5$$

$$\alpha_2 = 0$$

$$\alpha_3 = 0.5$$

Only non-zero alphas correspond to support vectors.

Support vectors:

$$A = (2, 2)$$

$$C = (2, 0)$$

Step 3: Compute Weight Vector

Using

$$\begin{aligned}w &= \sum_i \alpha_i y_i x_i \\&= 0.5(+1)(2, 2) + 0.5(-1)(2, 0) \\&= (1, 1) - (1, 0) \\&= (0, 1)\end{aligned}$$

Step 4: Compute Bias

Using support vector A:

$$y_i(w^T x_i + b) = 1$$

$$(+1) [(0, 1) \cdot (2, 2) + b] = 1$$

$$2 + b = 1$$

$$b = -1$$

Final Decision Function

$$\begin{aligned} f(x) &= w^T x + b \\ &= (0, 1) \cdot (x_1, x_2) - 1 \\ &= x_2 - 1 \end{aligned}$$

Classification rule:

$$\hat{y} = \text{sign}(x_2 - 1)$$

Prediction Example

New sample:

$$x = (3, 2)$$

$$f(x) = 2 - 1 = 1$$

$$\hat{y} = +1$$

Positive class.

What Have We Learned?

- Computed Gram Matrix
- Applied dual constraints
- Solved for alpha values
- Identified support vectors
- Computed weight vector w
- Computed bias b
- Built final classifier
- Predicted a new sample

This is the complete dual-form SVM workflow.